



Tool wear state recognition study based on an MTF and a vision transformer with a Kolmogorov-Arnold network

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ABSTRACT

In manufacturing, tool wear directly affects the quality, production cost and efficiency of machine tools. Accurately monitoring tool wear states is critical. However, 1D tool wear signal features are nonobvious. The existing models entail feature extraction processes that have difficulty considering long-sequence tool wear information in parallel and low robustness. Herein, an adaptive vision transformer (ViT) incorporating a Kolmogorov-Arnold network (KAN) based on a Markov transition field (MTF), MTF-AViTK, is proposed. The network can perform parallel processing and addressing long-term dependencies, better identifying tool wear states from cutting signals. A signal is converted into a 2D image via 2D time series encoding. The ViT performs token patch position encoding on 2D images and self-attention-based tool wear feature extraction in its encoder blocks. An adaptive multilayer perceptron (AdaptMLP) module is introduced to fine-tune the feature extraction process. The extracted features are mapped by the KAN to identify the tool wear state. PHM2010 provides two datasets for validating the performance and robustness of MTF-AViTK model. The highest model recognition accuracies are 95.76 % and 95.71 %. The results show that this model has better robustness and generalizability than the competing approaches. Finally, ablation experiments verify that each module has a positive effect on model performance. This method can accurately identify tool wear states based on machining data, enabling more accurate tool change decisions.

1. Introduction

Manufacturing has entered the digital age with advances in data collection systems, information technology (IT) and networks [1]. Computerized Numerical Control (CNC) machine tools are important production equipment in manufacturing, and the final execution unit is a cutting tool. Tool wear is a decisive factor that affects product quality, power consumption, costs and other issues. The tool wear problem has been at the forefront of research for many years [2]. For data-driven smart manufacturing processes, tool wear state monitoring technology has dramatically influenced the intelligent development of modern industries [3,4].

Promptly recognizing tool wear states can effectively guarantee machining quality. The methods for determining tool wear states are usually classified as direct or indirect methods [5]. Chen et al. [6] also describe the concept of existing surveillance in detail. A

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direct method is based on the operator's empirical judgment in a practical production case, and it is difficult to accurately identify tool wear states. And de Lacalle et al. [7] developed a diagnostic tool that allows researchers to detect potential milling problems and reduce the processing costs of complex parts. Aldekoa et al. [8] evidenced that no-sensor machine variable monitoring can achieve tool wear monitoring in broaching in real production environments. Bai et al. [9] presented a new digital twin-based process monitoring method and system in small batch machining, and the cutting model is integrated into the monitoring method to improve the diagnosis accuracy. Tapia et al. [10] studied the relationship between industrial communication protocols and monitoring data in industrial environment.

To prevent quality deteriorations, cutting tools are frequently replaced prematurely, which leads to significant waste. An indirect method involves processing signal data via sensors, which are processed by a data-driven model. A strong mapping relativity between the data and tool wear is reflected. Indirect methods have been extensively studied owing to their ability to accurately obtain tool wear states. Therefore, it is preferable to determine tool wear states on the basis of signals.

Due to the development of big data and artificial intelligence, smart sensors and high-precision measuring devices are increasingly being employed in production systems [11]. By collecting and analyzing multisource data from the machining process, researchers have found a relationship between this process and tool wear. The data were integrated and analyzed through modern technological methods to discover the pertinent mapping relationship between machining data (such as acceleration, strain, force and acoustic signals e.g.) and tool wear states. New advances in machine learning (ML) including deep learning (DL) during the last decade have fueled the rise of data-driven smart manufacturing [12]. To date, numerous scholars have conducted systematic research on tool wear states via machine learning [13] and deep learning [14].

Currently, numerous machine learning algorithms are applied to tool wear state recognition. Wei et al. [15] performed a spectral analysis on the milling force and established a force signal decomposition model based on whitened variational modal decomposition (WVMD). Joint information entropy (JIE) was used to acquire the features, and the optimal subset of features could reflect the mapping relationship between the features and the wear state. The ideal path forest algorithm was exploited to complete the classification task. Liu et al. [16] proposed an ellipsoidal network model based on an incremental learning algorithm combined with a fast correlated filtering method for online learning and tool wear state identification. Olmo et al. [17] proposed an efficient broaching monitoring method that combines real-time monitoring with tool wear detection. Xie et al. [18] predicted tool wear states via random forest (RF) and extreme gradient boosting (XGBoost) algorithms. Other scholars have conducted research on support vector machines (SVMs) [19], hidden Markov models (HMMs) [20], and relevance vector machines (RVMs) [21]. For the above algorithms, it is generally necessary to design feature engineering strategies on the basis of expert experience. Manual feature extraction and selection processes frequently miss some important features contained in the raw signal. Moreover, these algorithms have the disadvantages of shallow network structures and weak generalization capabilities. With the increasing amount of machining data, it is impractical for workers to manually extract features. In addition, traditional ML algorithms have insufficient generalizability for tool wear state monitoring, which leads to decreased performance.

Owing to their excellent adaptive feature extraction capabilities and powerful fitting performance, deep learning algorithms are widely applied in tool wear state monitoring [22]. Convolutional neural networks (CNNs) [23], recurrent neural networks (RNNs) [24], long short-term memory (LSTM) [25] and their hybrid networks have been commonly investigated for qualitatively and quantitatively monitoring tool wear. Deep learning techniques can adaptively extract relevant features from original signals, which enable the learning and establishment of complex mapping relationships [26]. Deep learning-based tool wear condition monitoring models typically extract 1D or 2D spatial features from machining signal sequences [27–30]. Marani et al. [31] proposed an LSTM network based on the motor current signal of a lathe spindle as its input. The tool wear value was predicted during the machining of steel alloys. Gudelek et al. [32] proposed a wavelet LSTM-deep multilayer perceptron (DMLP)-based model, which utilizes the continuous wavelet transform for preprocessing of raw data, LSTM for extracting temporal information, and DMLP for regression of the tool wear. Shu et al. [33] proposed a CNN-based self-supervised comparison learning framework that achieved accurate tool wear recognition. Sener et al. [34] presented a chatter detection method based on deep convolutional neural network (DCNN). Martinez-Arellano et al. [35] proposed a tool wear classification method based on signal imaging and a CNN. The manual feature selection process was avoided while the original data features were preserved. Research has been carried out on the use of LSTM, CNNs and their variants for tool wear monitoring. The performance of LSTM is satisfactory over inputs with limited range lengths, but it is unable to perform parallel computations. This affects the performance of LSTM when faced with longer signal input conditions [36]. While CNNs can be computed in parallel, increasing their acceptance fields requires the stacking of convolutional layers, which can lead to gradient vanishing or gradient explosion. Although network structure modification methods, such as ResNet and attention mechanisms, can improve their performance to some extent, their intrinsic architecture limit ability to establish long-distance time-series relationships [5]. Therefore, there is a need to construct an algorithm that not only operates in parallel but also considers the long-range temporal relationships of machining signals to improve the generalizability and robustness of tool wear monitoring.

The transformer architecture has recently attracted considerable attention because of its effectiveness in a range of domains, including language, vision and reinforcement learning [37]. The transformer architecture, which enables parallel computing and automatic feature extraction for longer time series data, was proposed by Vaswani et al. [38] in 2017. Transformers have demonstrated their strength in numerous fields, especially natural language processing [39], computer vision [40], and literature [41–44], and they have triggered technological breakthroughs such as ChatGPT. In various visualization benchmarks, transformer-based models perform similarly to or better than other networks, especially CNNs and RNNs [45]. In manufacturing cases, researchers have attempted to apply transformers to fault diagnosis and forecasting tasks. Ding et al. [46] developed a transformer model and applied it to bearing fault diagnosis. Liu et al. [47,48] first employed a transformer for tool wear prediction. They further proposed a CNN-Transformer model to achieve improved prediction performance. Li et al. [5] developed a tool wear forecasting model with a backbone

consisting of an Informer encoder and a stacked bidirectional gated recurrent unit (IE-SBiGRU) network. Xue et al. [49] improved the Swin transformer network to form a hierarchical time series transformer network (HTT-Net). The self-attention time series data calculation and the division window of this method simplify HTT-Net complexity. The model was validated with respect to both performance and robustness for tool wear classification and identification.

However, the current applications of transformers in tool wear state classification and prediction scenarios are based on 1D time series data. Studies based on 2D images have applied CNNs and their variants. Alexey et al. [50] proposed the vision transformer (ViT), proving that dependence on a CNN is unnecessary and that a pure transformer directly applied to image patch sequences can efficiently perform image classification tasks. Additionally, the ViT model was properly adapted to downstream tasks and manifested exciting results [51,52]. In recent years, ViT models have been extensively explored in the vision recognition field. Li et al. [53] proposed the vision Outlooker (VOLO) architecture, which implements a novel Outlook attention operation that can dynamically execute a local feature aggregation mechanism in the input image. Jing et al. [54] proposed a novel extended ViT (ExViT) model and verified that the ExViT outperformed converter-based methods and CNNs. Pan et al. [55] employed an end-to-end ViT-AMC network which integrates ViT and AMC modules. Chen et al. [56] proposed an effective transformer adaptation method, AdaptFormer, which can efficiently adapt pretrained ViTs to different images and video tasks.

ViTs have gained widespread attention for use in image classification tasks. Therefore, an innovative AViTK network architecture based on a 1D time series encoding technique, an MTF, is developed in this study. MTF-AViTK can perform tool wear recognition with long-term dependencies and global parallel operations.

The principal contributions of this study are summarized as follows.

(1) In this study, a novel AViTK network architecture is proposed, and a ViT-based model is applied to the tool wear state recognition task for the first time. It can adaptively process time series data and parallelize the feature extraction process. A mapping from the extracted features to the tool wear state is established.

(2) One-dimensional time series tool wear data are converted into MTF images for the first time. MTF encoding technique amplifies the local feature information from a 2D perspective, which can more intuitively characterize the signal changes caused by tool wear. This encoding technique can reflect the evolution of the data state. The information contained in the time series data is extracted via the proposed model.

(3) More nonlinear features are accepted by the ViT to enhance the nonlinear transformation of the feature vectors for each location patch. KAN is exploited to learn wear feature tokens and reduce the dimensionality of the model to prevent overfitting. Finally, Cross-validation and ablation experiments demonstrate that MTF-AViTK is better than or equivalent to the current methods.

The remainder of this article is organized as follows. The general architecture of MTF-AViTK is presented in Section 2, and the mechanism of each module is described. The PHM2010 dataset is preprocessed, and the training parameters are set in Section 3. The experimental outcomes are analyzed and discussed in Section 4. Conclusions are outlined in Section 5.

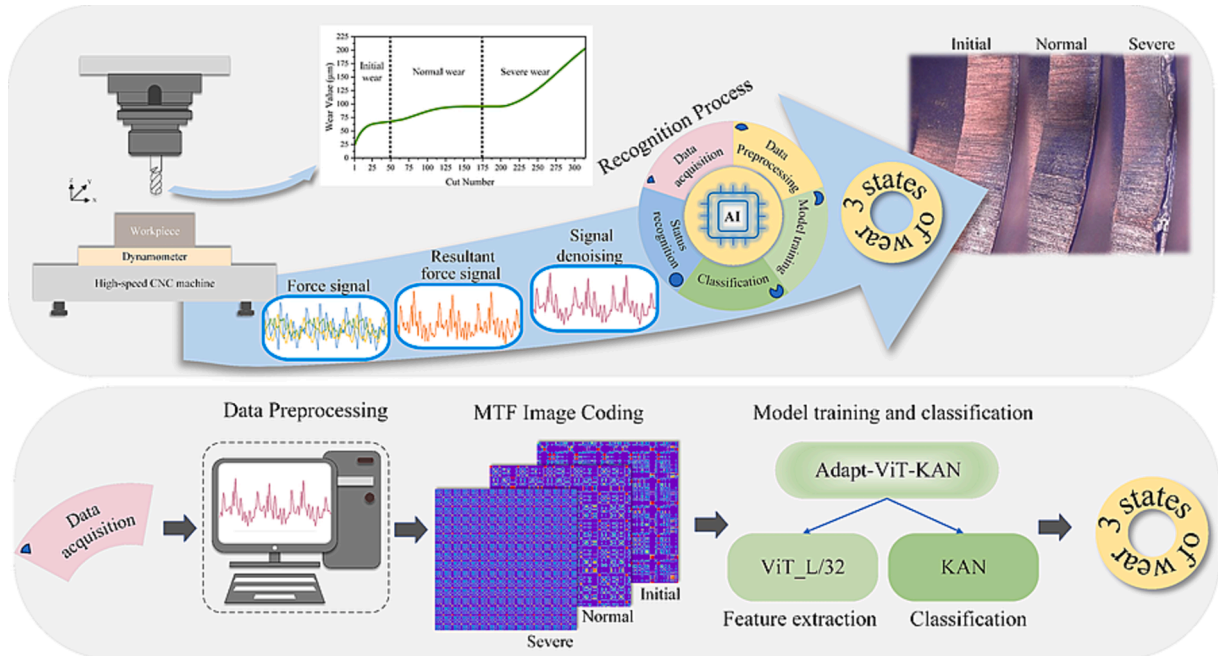


Fig. 1. Overall framework.

2. Tool wear state monitoring methodology based on MTF-AViTK

2.1. Overall framework

Fig. 1 shows the general framework of the tool wear state recognition method based on MTF-AViTK. Fig. 2 shows the network schematic of AViTK. The specific procedure is as follows.

(1) One-dimensional digital time series signals are collected during machining as the original signals. Moreover, the flank-face wear value of the cutting tool is measured at each cutting step.

(2) The acquired multichannel digital signals undergo dimensionality reduction and are synthesized into total digital 1D time series signals that correspond to each tool wear value.

(3) The processed signals are denoised via wavelet threshold denoising, and the denoised 1D time series signal is converted to a 2D image via the MTF image encoding technique. The MTF 2D images are divided into three datasets on the basis of their tool wear states: initial wear, normal wear and severe wear.

(4) The obtained triple tool wear classification datasets are used to train and verify the AViTK model. Adapt-ViT_L/32 is used to extract the features of 2D image data. The extracted features are classified and mapped by a KAN [57] to finally obtain the tool wear states.

2.2. MTF image encoding technique

The 1D time series signals acquired during machining are encoded into 2D image data. The MTF encoding process is as follows.

First, all the sampling points of the signal data $X = \{x_1, x_2, \dots, x_i, x_j, \dots, x_n\}$ are normalized to the interval $[0, 1]$, and new data $X' = \{x'_1, x'_2, \dots, x'_i, x'_j, \dots, x'_n\}$ are obtained. Second, quantile bins are created based on the range of values contained in the time series X' . The number of quantile bins is Q , and the bins are denoted by q_i and q_j ($q \in [1, Q]$), which contain time stamps i and j (temporal axis), respectively. Each data quartile bin belongs to a unique interval, and each x'_i is assigned to the corresponding square, which is known as q_j ($j \in [1, Q]$). Third, the transitions between quantile bins in the form of first-order Markov chains along the time axis are computed, constructing a weighted adjacency matrix with a size of $[Q, Q]$ called W . Finally, W_{ij} is determined by the frequency of a point in the quantile bin q_j being followed by q_i . W is a Markov transfer matrix. To compensate for the loss of information, W is converted into an MTF matrix with a size of $[n, n]$, which is denoted as M . Eq. 1 defines the MTF.

$$M = \begin{bmatrix} W_{ij|x'_1 \in q_i, x'_1 \in q_j} & W_{ij|x'_1 \in q_i, x'_2 \in q_j} & \dots & W_{ij|x'_1 \in q_i, x'_n \in q_j} \\ W_{ij|x'_2 \in q_i, x'_1 \in q_j} & W_{ij|x'_2 \in q_i, x'_2 \in q_j} & \dots & W_{ij|x'_2 \in q_i, x'_n \in q_j} \\ \dots & \dots & \dots & \dots \\ W_{ij|x'_n \in q_i, x'_1 \in q_j} & W_{ij|x'_n \in q_i, x'_2 \in q_j} & \dots & W_{ij|x'_n \in q_i, x'_n \in q_j} \end{bmatrix} \quad (1)$$

where W_{ij} is the jump probability of $q_i \rightarrow q_j$. The matrix W containing the transfer probabilities on the magnitude axis is expanded to form an MTF matrix considering the time position.

2.3. Adapt-ViT_L/32 network

MTF image features are extracted by the Adapt-ViT_L/32 network. The main structure of the Adapt-ViT_L/32 network is divided into two modules: ViT_L/32 and AdaptMLP. The architecture of the MTF-AViTK is shown in Fig. 3.

The ViT_L/32 model can be divided into three parts: a positional coding module, a multilayer sequential self-attention encoder, and a multiclassification layer. First, the MTF-encoded images are converted to the input format $(384 \times 384 \times 3)$ in the input layer. The images subsequently undergo position encoding. Each image $x \in R^{H \times W \times c}$ is divided into consecutive $x_e \in R^{N \times (P^2 d)}$ patches after the patch embedding layer. Here, (H, W, c) represents the *height*, *width* and *number of channels* contained in the input MTF image; (P, P)

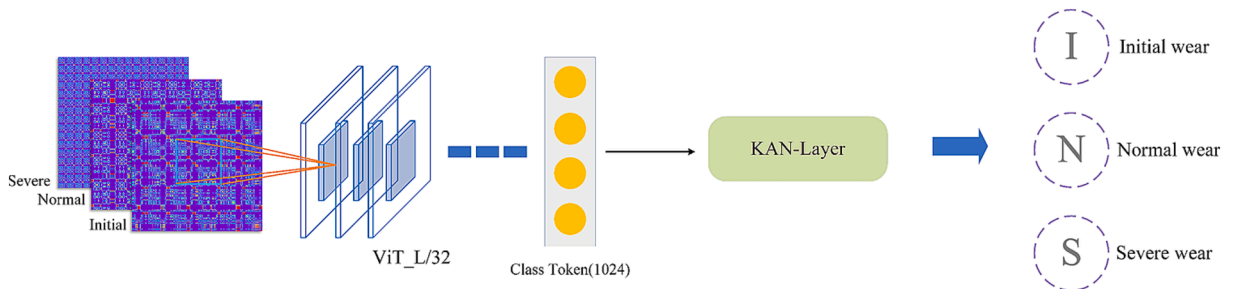


Fig. 2. Architecture of AViTK.

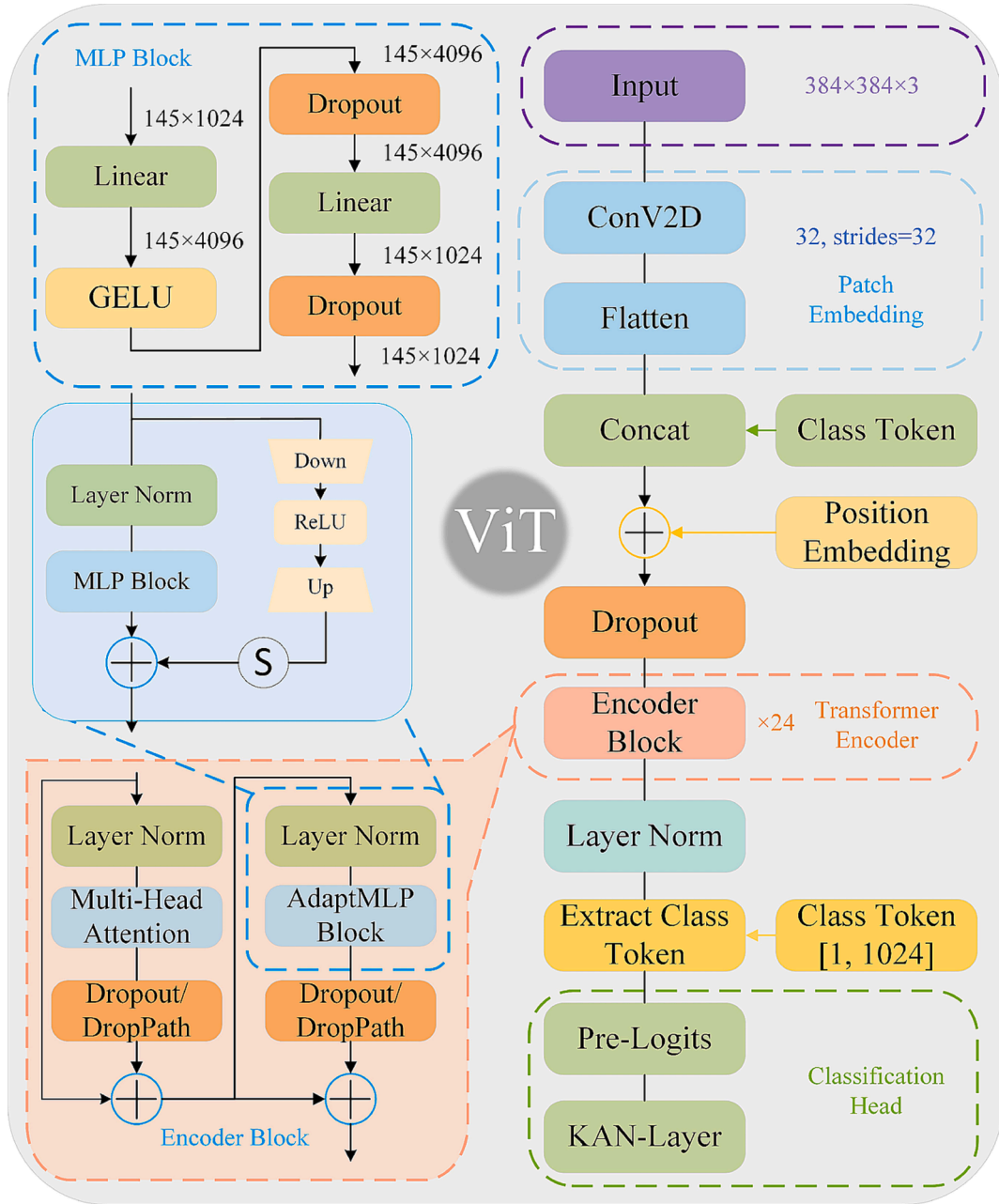


Fig. 3. Structure of Adapt-ViT_L/32.

signifies the resolution of each image patch; and d denotes the dimensionality of the output. The number of position-encoded tokens (in the ViT model, the split patches are called “tokens”) is calculated via Eq. 2.

$$N = \frac{H \times W}{P \times P} \quad (2)$$

The position encoding module of ViT_L/32 converts the input image with sizes of $[384, 384, 3]$ into $[12, 12, 1024]$ by a 2D convolutional kernel with a size of 32×32 and a step size of 32. Subsequently, the image is flattened into a 2D matrix with a size of $[144, 1024]$. This 2D matrix is entered to the encoder, a special trainable parameter vector called x_{class} ([class] token) and a position embedding x_{pos} for classification must be embedded. After the [class] token is spliced, the 2D matrix becomes $[145, 1024]$. The linear embedding process can be expressed as Eq. 3. Taking 9 patches as examples, a sketch of the positional encoding process is shown in Fig. 4.

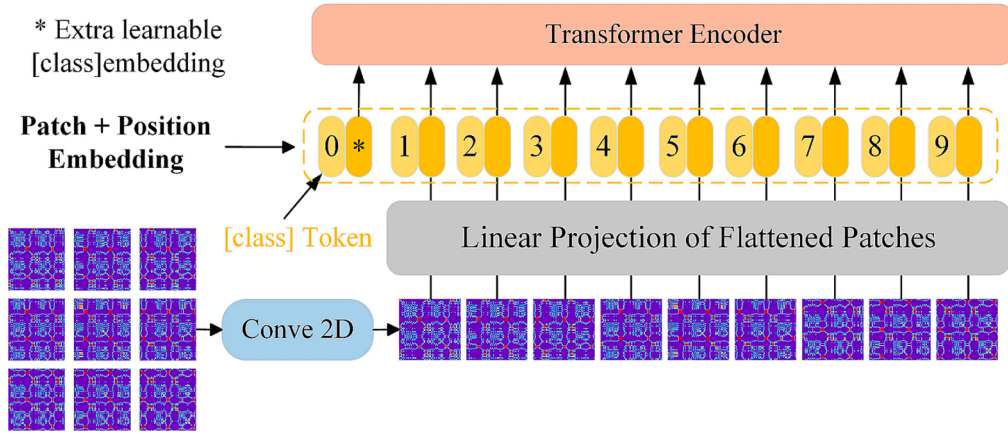


Fig. 4. Position encoding process.

$$\mathbf{x}_0 = [\mathbf{x}_{class}; \mathbf{x}_e] + \mathbf{x}_{pos}, \mathbf{x}_{pos} \in \mathbb{R}^{(n+1) \times d} \quad (3)$$

After completing positional encoding, the data are input into a multilayer self-attention transformer encoder. Each encoder is composed of a multi-head self-attention (MHSA) layer and an AdaptMLP layer. The MHSA layer is constructed from multiple self-attention mechanisms. It can be expressed by Eqs. 4 and 5 [58]. The details of this process are given in the [Appendix](#).

$$\mathbf{S}' = \text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q} \cdot \mathbf{K}^T}{\sqrt{d_k}}\right) \mathbf{V} \quad (4)$$

$$\mathbf{S}_l'' = \text{MultiHead}(\mathbf{Q}', \mathbf{K}', \mathbf{V}') = \text{Concat}(h_1, \dots, h_h) \mathbf{W}^O \quad (5)$$

where $\mathbf{S}'$ is a weighted sequence. Normalization and gradient vanishing mitigation are achieved by dividing $\sqrt{d_k}$ after the scaling step. $\mathbf{S}_l''$ is the token produced by the MHSA mechanism at layer l . h_i denotes each head of the self-attention mechanism (in this study, this value is set to 16), where the attention is $(\mathbf{Q}_i, \mathbf{K}_i, \mathbf{V}_i)$. Each head has a different matrix for the query vector $\{\mathbf{Q}_i\}_{i=1}^m$, key vector $\{\mathbf{K}_i\}_{i=1}^m$, and value vector $\{\mathbf{V}_i\}_{i=1}^m$. $\mathbf{W}^O$ is a linear projection matrix.

The output tokens are subsequently sent to layer normalization (LayerNorm) and AdaptMLP blocks via the MHSA mechanism. The AdaptMLP block consists of two branches. The MLP layer in the left branch is the same as that in the original network, as shown in the MLP block presented in Fig. 3. The MLP block consists of two fully connected layers in which a Gaussian error linear unit (GELU) is added. The right branch is an additional lightweight module introduced for task-specific fine-tuning [56]. The AdaptMLP process can be expressed as Eq. 6.

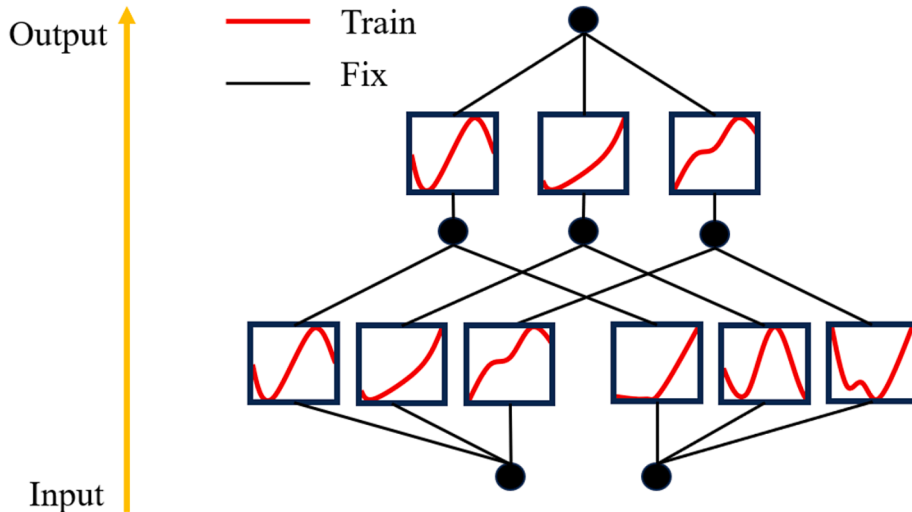


Fig. 5. KAN.

$$S_l = \text{MLP}(\text{LayerNorm}(S_l'')) + s \cdot [\text{ReLU}(\text{LayerNorm}(S_l'') \cdot W_{\text{down}}) \cdot W_{\text{up}}] S_l'' \quad (6)$$

where S_l is the output of the l -th layer encoding block and S'' represents the adapted features. $W_{\text{down}} \in R^{d \times \hat{d}}$ is a downward-projecting layer, and $W_{\text{up}} \in R^{\hat{d} \times d}$ is an upward-projecting layer. $\hat{d}$ is the bottleneck intermediate dimensionality that satisfies $\hat{d} \ll d$. s is the scale factor of the residuals.

In this study, an input 2D matrix with a size of [145,1024] is output as [145,4096] through the first linear layer of the MLP block. The second linear layer restores the size of the output to [145,1024]. Each input vector of the encoder block is mapped to an output vector. Only the [class] token, which corresponds to the encoder block, is taken with a size of [1,1024]. This means that [1, 1024] (corresponding to the [class] token) is extracted from [145,1024]. The final classification result is subsequently obtained by the classification head.

2.4. KAN

A general KAN is a combination of L layers. Given an input vector $x_0 \in R^{n_0}$, the output of a KAN is as shown in Eq. 7. The details of the related process are given in the [Appendix](#).

$$\text{KAN}(x) = (\Phi_{L-1} \circ \Phi_{L-2} \circ \dots \circ \Phi_1 \circ \Phi_0)x \quad (7)$$

KANs treat linear transformations and nonlinearities together in Φ [57]. In this study, a diagram (Fig. 5) is drawn for the KAN (three layers are used as an example).

The last linear fully connected layer of the original ViT architecture is replaced by a KAN for mapping the network feature classification process after performing extraction. Here, a two-layer KAN structure is used. The input is the feature dimensionality, which is the [class] token corresponding to [1,1024] extracted from [145,1024]. The output is the three-class classification of the corresponding tool wear states. Table 1 shows the hyperparameters selected for the KAN.

3. Data processing and parameter settings

3.1. Data description

The PHM2010 dataset [59] is used to validate the performance of the proposed MTF-AViTK tool wear state recognition model. This dataset measures the full life cycle of the 6 cutting tools with 3-flute carbide ball-end milling cutters, which are denoted as C1-C6. Here, experimental cutting tools (C1, C4 and C6) record the wear values for each cutting pass. The workpiece material was processed by these three cutting tools on a high-speed milling machine, and the machining conditions were fixed-parameter dry milling. The specific information is shown in Fig. 6. The detailed machining parameters are shown in Table 2.

The distance between each cutting pass was 108 mm. As shown in Fig. 6, the dataset was acquired via a Kistler 3-term force dynamometer to obtain the cutting force signals. Vibration signals was acquired via a Kistler 3-term piezo accelerometers. Moreover, a Kistler acoustic emission (AE) device was installed to acquire AE signal. The dataset includes these three types of signals, where the force and vibration signals are three-channel signals, in which three axes (x, y, z) are included. The AE signal is a single-channel signal. These seven signals were captured by DAQ NI PCI 1200 at a sampling rate of 50 kHz/channel. After each cutting pass, the flank wear width of each tooth was measured offline via a Leica MZ12 microscope as the tool wear value. A total of 315 sets of experimental data were collected for each cutting tool. The wear trend of the 3-flute cutter of C1 is visualized in Fig. 7(a). The wear width of each tooth is different and nonlinear. Judging the overall wear of a cutting tool by the wear value of a single tooth is not accurate. The average of the 3-flute wear values was acquired as the objective in this study. The wear curves for the life cycles of the cutting tools are shown in Fig. 7(b). The tool wear phenomenon can be classified into three different stages: initial wear, normal wear and severe wear.

3.2. Dataset preprocessing and signal denoising

The original force signals of experimental cutting tools were inputted into the model. The signals acquired from the stable cutting process for each cutting pass were selected as the signal window: 90,000 ~ 120,000. A data window that is too large leads to a computational cost increase. Therefore, a data window with a length of 10,000 (90,000 ~ 100,000) was selected as the stable sampling window for each cutting pass. A nonoverlapping window with a length of 2000 was used to divide the data of each cutting pass into five single samples with sizes of $2000 \times (x,y,z)$. The force signals of the three directions in each sample were synthesized into the resultant

Table 1
Settings of the main hyperparameters in the KAN.

Parameters	Values
The number of grid intervals (G)	5
The order of a piecewise polynomial (k)	3
Residual function b(x)	SiLU(x)
The scale of the residual function b(x) and the base function spline(x), which are trainable	1.0

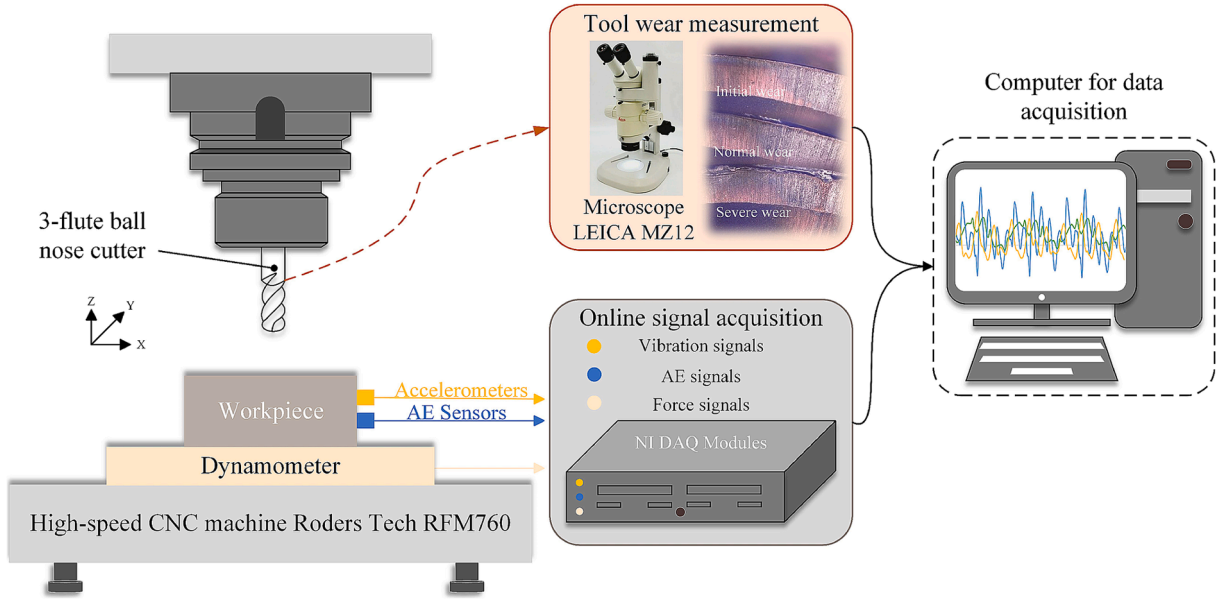


Fig. 6. Schematic of the PHM2010 data acquisition process.

Table 2
Experimental parameters used for PHM2010.

Experimental settings	Parameters
Workpiece material	Inconel 718 alloy
Spindle speed (n)	10400 rpm
Feed rate (f)	0.15 mm/r
Cutting depth (a_p)	0.2 mm
Cutting width (a_e)	0.125 mm
Sample rate	50 kHz

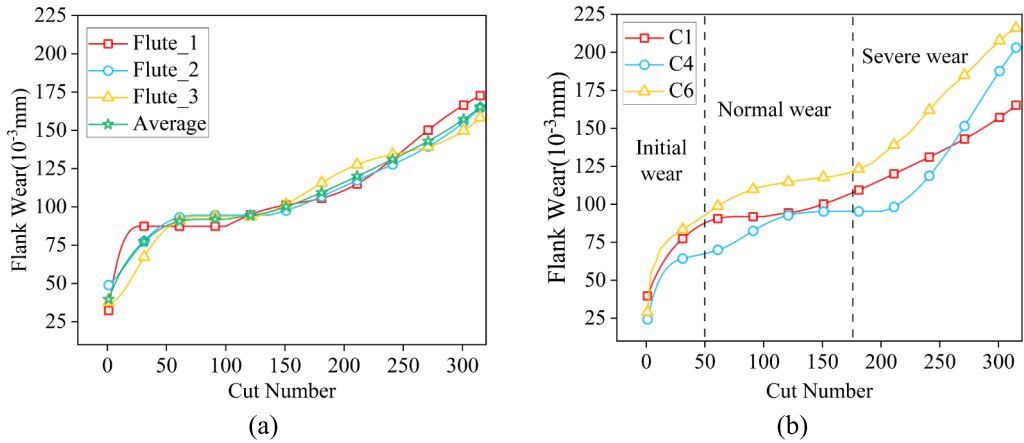


Fig. 7. Wear curves. (a) 3-flute tool wear curves of C1. (b) Tool wear curves of experimental cutting tools.

force signal via Eq. 8.

$$F = \sqrt{F_x^2 + F_y^2 + F_z^2} \quad (8)$$

where F represents the resultant cutting force. F_x , F_y , and F_z represent the force signals in the X , Y , and Z directions, respectively.

In practical processing cases, many noise signals are collected by the utilized sensor. Noise signal processing can better reflect the

characteristics of the signal changes caused by tool wear. The signal obtained after denoising can more accurately reflect the inherent characteristics. Hence, wavelet threshold denoising was adopted for filtering and denoising the cutting force signals, as discussed by Cheng et al. [60]. The wavelet thresholding denoising method can effectively preserve the features contained in the original signal. This method is well suited for analyzing non-smooth signals and extracting local features of the signal. Sym7 was used as the wavelet basis function. The decomposition scale was 6, and the wavelet soft thresholding method was used for denoising.

Sym7 wavelet, a member of the Symlets family, is designed to address the limitations of classical wavelets like the Daubechies wavelets. Sym7 wavelet finds extensive applications in signal processing, image analysis, and data compression owing to its balanced trade-offs between near-symmetry (Sym7 approximates symmetry better than Daubechies wavelets, reducing phase distortion during signal reconstruction), high vanishing moments (Captures abrupt changes while ignoring smooth polynomial trends in signals), and compact support (Operates within a finite interval, enabling localized time–frequency analysis).

The core characteristic of Sym7 wavelet is its discrete nature, featuring 7 vanishing moments and 14 filter coefficients. It is part of the Symlets family, optimized by Ingrid Daubechies to improve symmetry while retaining key properties of Daubechies wavelets. The scaling function (father wavelet) and wavelet function (mother wavelet) of Sym7 are implicitly defined through filter coefficients and two-scale equations. The mathematical essence of Sym7 resides in its low-pass filter (decomposition scaling coefficients) and high-pass filter (decomposition wavelet coefficients), which govern the wavelet decomposition and reconstruction processes. Notably, Sym7 possesses seven vanishing moments, indicating its enhanced capability to effectively capture higher-order discontinuities in signals, such as abrupt singularities. The mathematical formulations are presented as follows:

(1) Vanishing Moments:

A wavelet has p vanishing moments which can be denoted as Eq. (9):

$$\int_{-\infty}^{\infty} t^m \psi(t) dt = 0 \quad m = 0, 1, \dots, p-1 \quad (9)$$

where Sym7's value is $p = 7$. This implies that Sym7 wavelet enhances sparsity in signal representations, critical for compression and denoising.

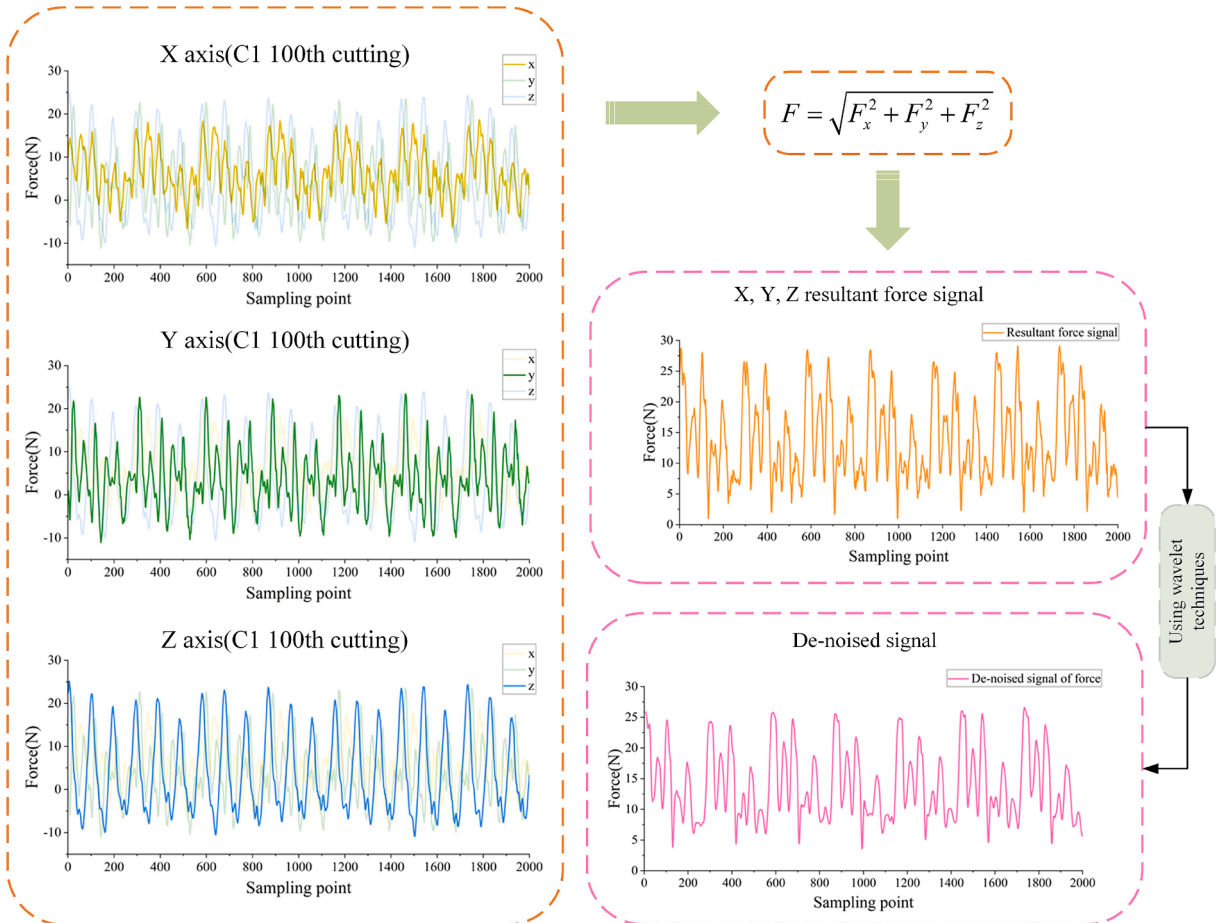


Fig. 8. Preprocessing sequence diagram for the 100th force signal of C1.

(2) Two-Scale Equations:

The scaling function Eq. (10) and wavelet function Eq. (11) are recursively defined via filter coefficients:

$$\varphi(t) = \sqrt{2} \sum_{k=0}^{13} h_k \varphi(2t - k) \quad k = 0, 1, \dots, 13 \quad (10)$$

$$\psi(t) = \sqrt{2} \sum_{k=0}^{13} g_k \psi(2t - k) \quad k = 0, 1, \dots, 13 \quad (11)$$

where h_k and $g_k = (-1)^k h_{13-k}$ denote the low-pass and high-pass filter coefficients, respectively. The high-pass and low-pass filters both possess filter coefficients of length 14. The approximate symmetry in the filter coefficients ($h_k \approx h_{13-k}$) serves to minimize phase distortion during signal processing. Since explicit analytical solutions for $\varphi(t)$ and $\psi(t)$ are unavailable, these functions are numerically approximated via iterative algorithms such as the cascade algorithm. This algorithm recursively refines initial approximations of $\varphi(t)$ and $\psi(t)$ by iteratively applying the two-scale equations, ultimately converging to their stable numerical representations.

The Sym7 of primary functions include denoising, compression, feature extraction, and multi-resolution analysis across domains. By minimizing phase distortion and efficiently capturing signal discontinuities, Sym7 is particularly suited for applications requiring high-fidelity reconstruction and localized analysis. Consequently, this study employs the Sym7 wavelet to preprocess collected cutting force signals. The algorithm's multiresolution analysis capability, combined with its 7 vanishing moments, ensures precise localization of transient features and efficient noise suppression while preserving critical morphological characteristics of the original signal.

Taking the 100th cutting data point of C1 as an example, the force signal preprocessing sequence is given in Fig. 8.

In this study, the 1st ~ 50th data was classified as initial phase, the tool wear of the 51st ~ 175th data was classified as normal phase, and the tool wear of the 176th ~ 315th data was classified as severe phase. However, the distribution of the data for the three stages is unbalanced in Fig. 7(b). The volume of normal and severe phases data was much more than that for initial phase. Severe wear was 2.8 times more frequent than initial wear and 1.12 times more frequent than normal wear. To balance the data distribution, the initial wear data was up-sampled in a new partition window: 100,000 ~ 120,000. The normal wear down-sampled was randomly 100,000 ~ 120,000 times. Finally, the force signal data of C1, C4 and C6 were converted into 2D images ($500 \times 500 \times 3$) via the MTF. Experimental cutting tools contain three types of tool wear datasets, each containing 700 MTF image samples. Three classification image datasets, experimental cutting tools, are shown in Fig. 9. Two types of cross-validation datasets (A and B datasets) were set up to evaluate the performance of MTF-AVITK. The details of the settings of the A and B datasets are shown in Table 3 and Table 4.

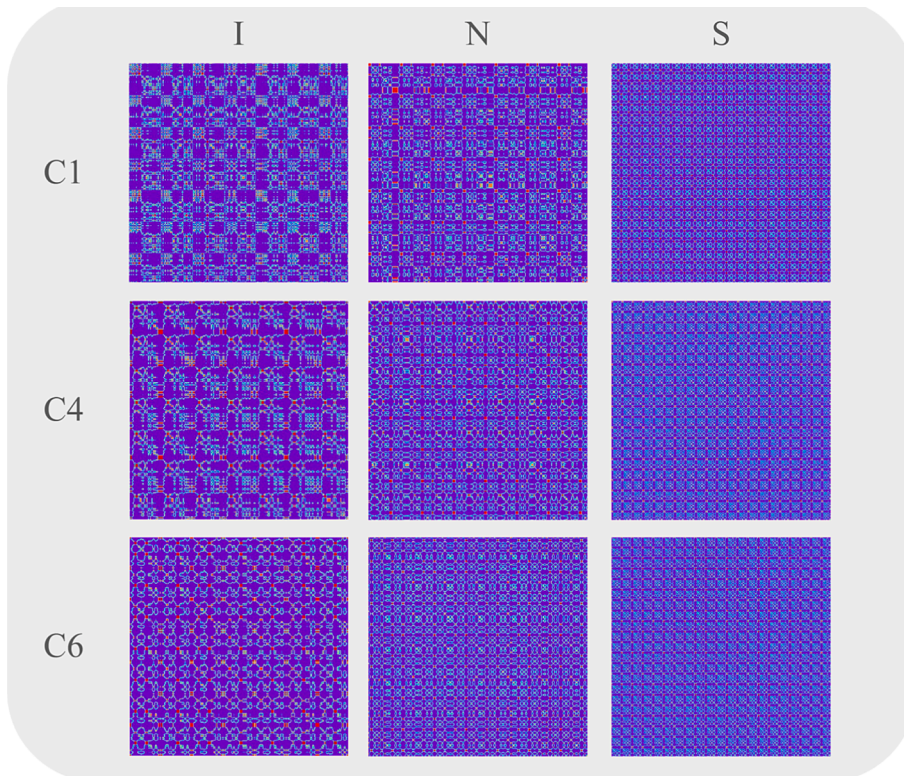


Fig. 9. MTF images of the experimental cutting tools datasets.

Table 3

The experimental division of the A datasets.

Datasets	Train	First test	Second test
A1	C1	A11 (Represents C4)	A12 (Represents C6)
A2	C4	A21 (Represents C1)	A22 (Represents C6)
A3	C6	A31 (Represents C1)	A32 (Represents C4)

Table 4

The experimental division of the B datasets.

Datasets	Train	Test
B1	C1 and C4	C6
B2	C4 and C6	C1
B3	C6 and C1	C4

3.3. Training configuration

The server used for this study was configured with a Windows 11 operating system, 32 GB of RAM, an NVIDIA 4080 Super GPU (10240 CUDA cores, with a driver version of 552.44), and the CPU was a 13th Generation Core i7 at 3.40 GHz. The software versions used were PyCharm 2021.2.4 (Professional), PyTorch 2.2.0 + cu118, Anaconda3 2021.11, and CUDA version 12.4.131. The stochastic gradient descent (SGD) optimizer was used to iteratively optimize the MTF-AViTK training parameters. Moreover, a small batch strategy was used to train the model. The weight decay parameter method was employed to suppress overfitting. Momentum coefficients were introduced to accelerate the training process. The specific hyperparameter settings are given in [Table 5](#).

4. Results and discussion

4.1. Model training results

The model proposed in this study was trained according to the configuration described in [Section 3.3](#). The training processes utilizing the A and B datasets are visualized in [Figs. 10 and 11](#), respectively. The accuracy and loss curves produced on the B datasets are shown in [Fig. 10](#). The accuracy and loss curves produced on the A datasets are shown in [Fig. 11](#). The proposed model converged before epoch 15. The high accuracy obtained by the model trained on both datasets indicates that the model can effectively learn the key features in 2D encoded tool wear images.

4.2. Model performance evaluation

For the classification problem, a confusion matrix was used to visualize the performance of MTF-AViTK. [Figs. 12 and 13](#) show the confusion matrices yielded the proposed tool wear state recognition model for the A and B datasets, respectively.

In tool wear state recognition tasks, a model usually needs to reduce false alarms, which cause wastage due to the execution premature tool changes in the first two phases.

The severe phase should be the focus for all cutting tools to avoid tool changes that are too late, which can lead to machining quality degradation [49]. Therefore, the precision of the model classification results is more important in the first two phases. The recall, specificity and F1 score should be given more importance in the severe phase. The performance evaluation indicators obtained on the A and B datasets are given in [Table 6](#) and [Table 7](#), respectively. Comparisons between the evaluation metrics achieved by MTF-AViTK on the two datasets are given in [Fig. 14](#). The A datasets yielded average precisions of 90.73 % and 91.51 % for the initial and normal phases, respectively. The average recall, specificity, and F1 score were 94.82 %, 97.93 %, and 95.24 %, respectively, in the severe wear phase. The B dataset provided average precisions of 95.16 % and 92.73 % in the initial phase and normal phase, respectively. The average recall, specificity and F1 score in the severe phase were 95.13 %, 98.16 % and 95.94 %, respectively.

4.3. Experimental results comparison

The performance of MTF- AViTK was compared with that of the existing methods. The comparison was divided into two groups of datasets: A and B. However, compared with the A datasets, the B datasets have greater distributional differences between their training and validation datasets. Hence, the A datasets require more generalizability and better model robustness. Only Ref. [35] and Ref. [49]

Table 5

Experimental hyperparameter settings.

Hyperparameters	Learning rate	Weight decay	Number of epochs	Batch size	Momentum
Values	0.0006	2E-4	50	8	0.9

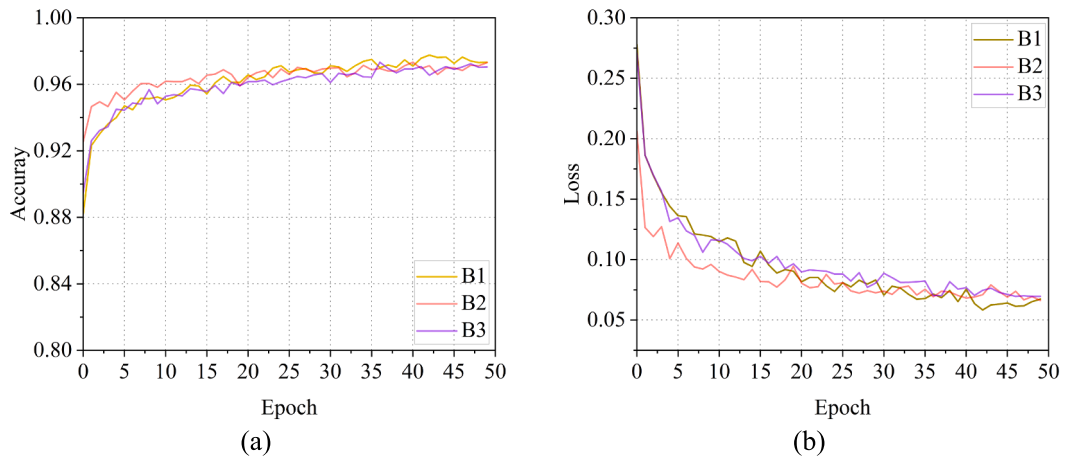


Fig. 10. Training accuracy and loss curves produced by MTF-AViTK on the B datasets. (a) Accuracy curves produced for the training datasets. (b) Loss curves produced for the training datasets.

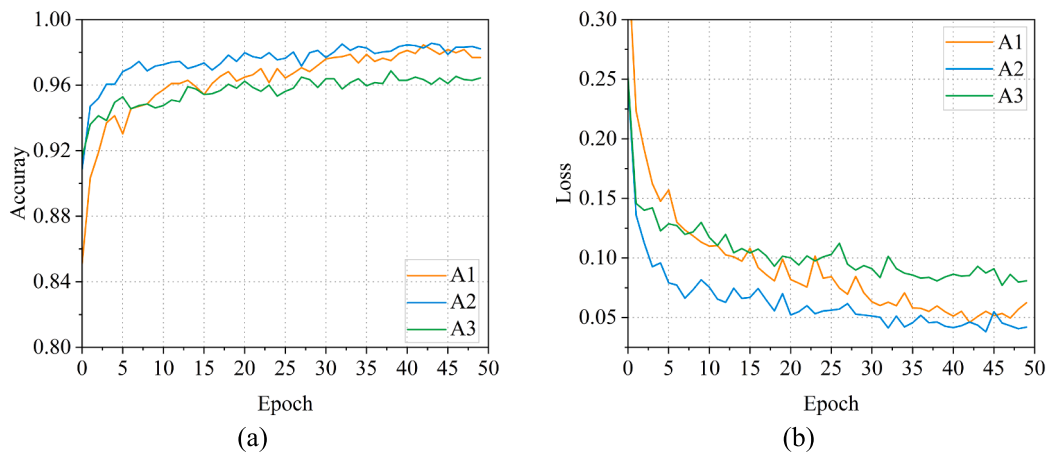


Fig. 11. Training accuracy and loss curves produced by MTF-AViTK on the A datasets. (a) Accuracy curves produced for the training datasets. (b) Loss curves produced for the training datasets.

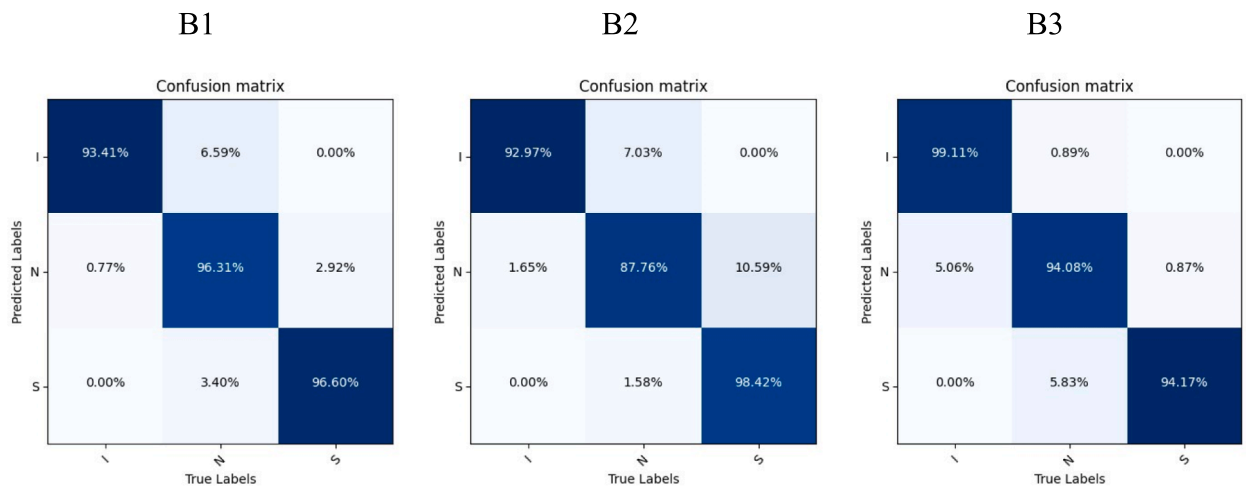


Fig. 12. Confusion matrices produced for the B datasets.

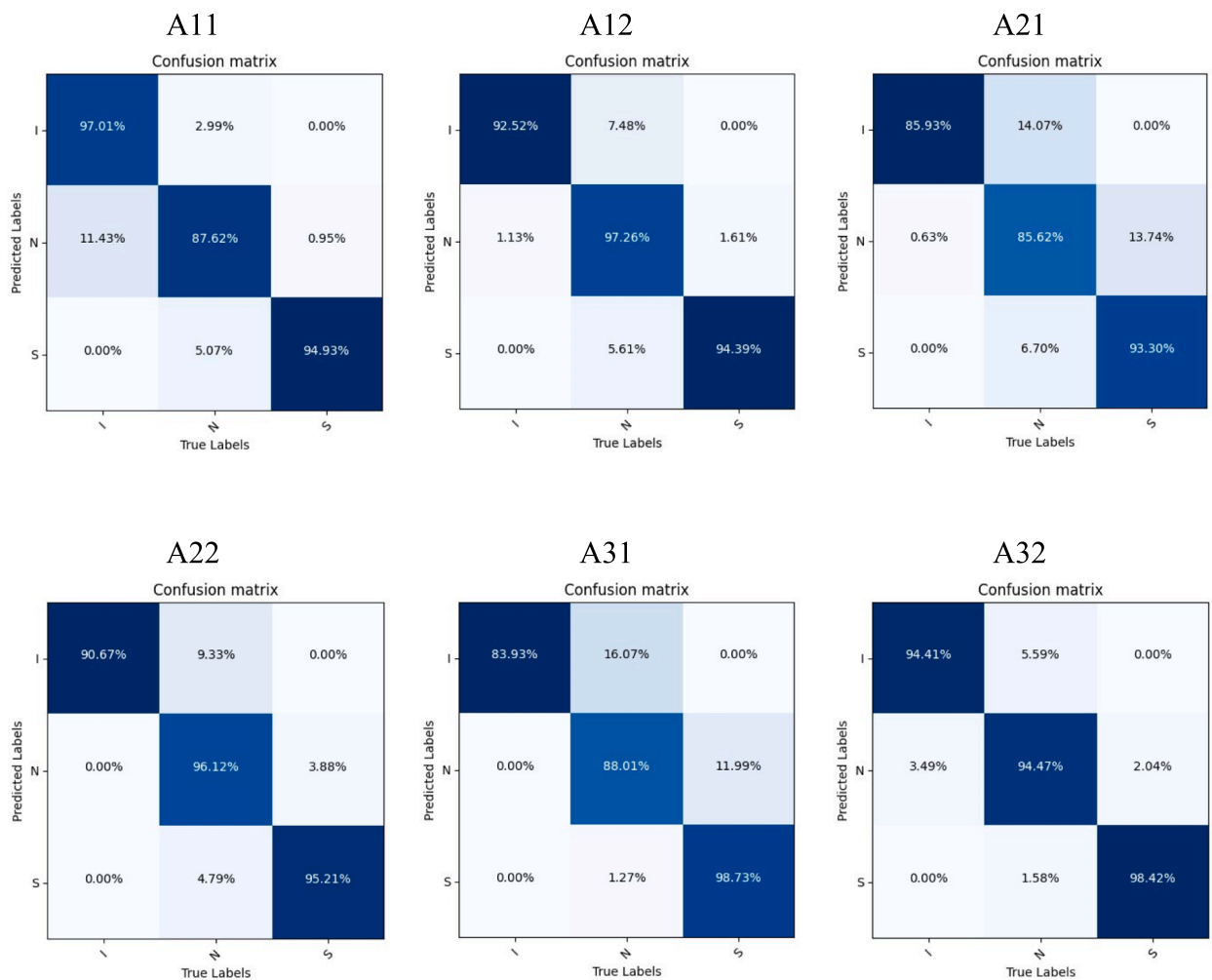


Fig. 13. Confusion matrices produced for the A datasets.

Table 6

Experimental indicators obtained on the A datasets.

Evaluation indicators		A11	A12	A21	A22	A31	A32
Accuracy (%)		94.57	93	88.14	93.81	89.62	95.76
	P _{initial}	92.5	97	85.9	90.7	83.9	94.4
Precision (%)	P _{normal}	97.3	87.6	85.6	96.1	88	94.5
	P _{severe}	98.6	99	87.6	96.6	89.1	98
Recall (%)	R _{severe}	97.1	97.4	96.9	97.6	99.4	99.2
Specificity (%)	S _{severe}	96.9	96.45	90.36	95.89	93.65	98.19
F1 score (%)	F _{severe}						

Table 7

Experimental indicators obtained on the B datasets.

Evaluation indicators		B1	B2	B3
Accuracy (%)		95.38	92.81	95.71
	P _{initial}	93.4	93	99.1
Precision (%)	P _{normal}	96.3	87.8	94.1
	P _{severe}	97.3	89	99.1
Recall (%)	R _{severe}	98.3	99.3	96.9
Specificity (%)	S _{severe}	97.79	93.46	96.59
F1 score (%)	F _{severe}			

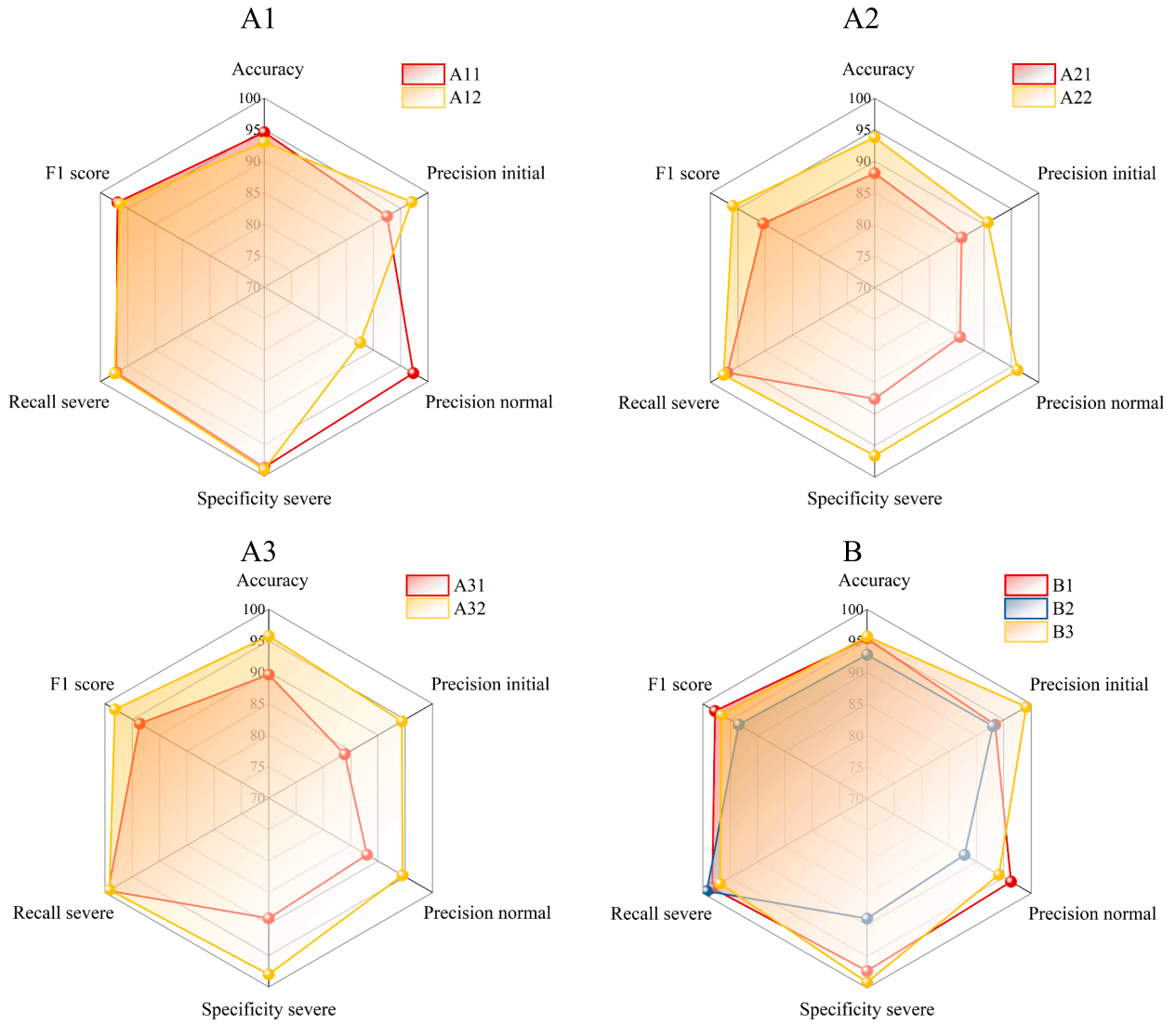


Fig. 14. Comparison between the evaluation indicators obtained on the A and B datasets.

performed relevant work on the A datasets among the existing studies. The model proposed in Ref. [35] uses a Gramian angular field (GAF) method to convert time series data into 2D images, which extract features through a CNN to classify tool wear states. Ref. [49] proposed HTT-Net for implementing tool wear state classification using 1D signals. The performance achieved by the model proposed in this study on the A datasets was compared with those of the above two methods, as shown in Fig. 15.

The accuracy of MTF-AViTK was greater than or equal to that of the existing methods, except that it performed slightly worse than HTT-Net did on the A11 dataset. In conclusion, the performance of MTF-AViTK was excellent on the A datasets, which further demonstrates the strong generalizability and robustness of this model.

Most researchers have used the B datasets to validate the performance of their models. This study compared the methods presented by many researchers, including machine learning, 1D CNNs, 2D CNNs, LSTM, transformers and their variants [61–64]. The classification performances of these models and the proposed model are compared in Fig. 16. The experimental results of HTT-Net and MTF-AViTK obtained on the B datasets are given in Table 8.

The classification accuracies of the proposed model reached 95.38 %, 92.81 % and 95.71 % on the B1, B2 and B3 datasets, respectively. In Fig. 16, except for the B3 dataset, the proposed model obtained almost the highest classification results among the above models. The best results were obtained by HTT-Net for the B3 dataset. In summary, the proposed tool wear state recognition model achieved better results on the PHM2010 dataset, making it highly superior to the existing methods. Moreover, the proposed model has very high generalizability and good robustness.

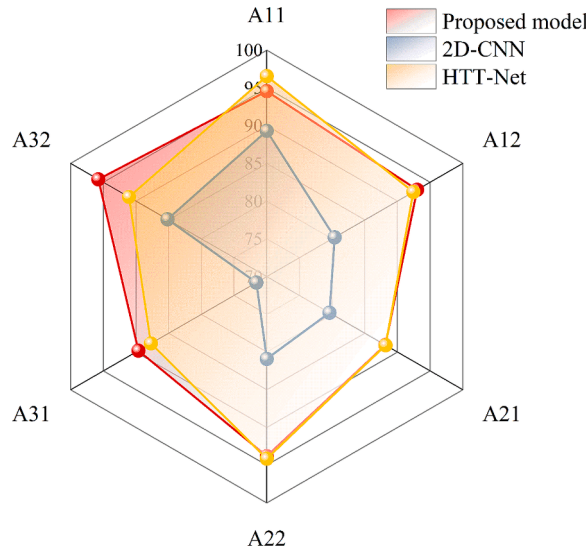


Fig. 15. Comparison among the model performances of the existing methods and the proposed approach on the A datasets.

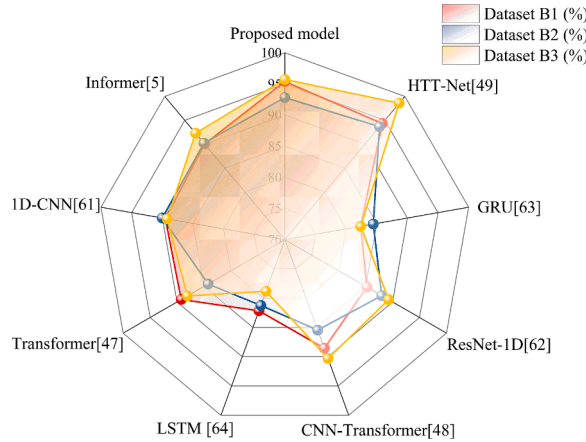


Fig. 16. Comparison among the model performances of the existing study and the MTF-AViTK approach on the B datasets.

Table 8

Comparison among the experimental results obtained on the B datasets.

Evaluation indicators	Dataset B1 (%)	Dataset B2 (%)	Dataset B3 (%)
HTT-Net [40]	94.35	93.67	98.59
Proposed model	95.38	92.81	95.71

4.4. Ablation experiments

In the ablation experiments, three models (Model 1, Model 2 and Model 3) were designed to validate the contributions of the AdaptMLP and KAN-Layer modules to the performance of MTF-AViTK. The AdaptMLP was eliminated in Model 1. The KAN-Layer was replaced by a traditional fully connected layer (MLP) in Model 2. Model 3 is the original network, which is a network directly converted from ViT_L/32. The B datasets were utilized to evaluate the contribution of the AdaptMLP and KAN-Layer module. The performance comparison among MTF-AViTK and its variants in the ablation experiment is shown in Fig. 17. A comparison of the experimental results is given in Table 9.

The ablation experiments demonstrate that the classification performances of the models in which the relevant modules were eliminated all degraded. Compared with Models 1, 2 and 3, the proposed AViTK model yielded accuracy improvements of 0.1 %, 0.34 % and 0.86 %, respectively, on the B1 dataset. It provided accuracy enhancements of 1.96 %, 1.2 % and 1.53 % on the T2 dataset over

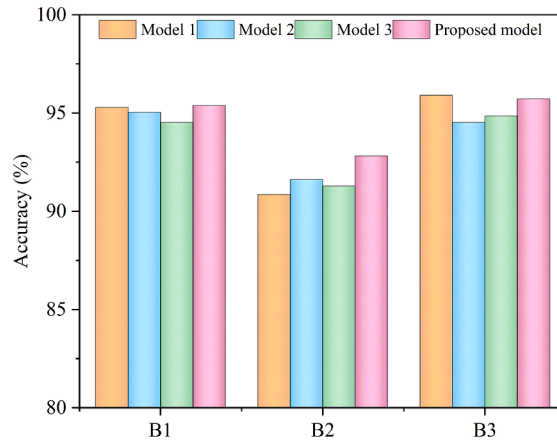


Fig. 17. Comparison among the accuracies in ablation experiments.

Table 9

Results of the ablation experiments.

Models	Dataset B1 (%)	Dataset B2 (%)	Dataset B3 (%)
Model 1 (no AdaptMLP)	95.28	90.85	95.90
Model 2 (no KAN-Layer)	95.04	91.61	94.52
Model 3 (ViT_L/32)	94.52	91.28	94.85
Proposed model	95.38	92.81	95.71

Models 1, 2 and 3, respectively. In addition, it improves the accuracy values by 1.19 % and 0.86 % on the B3 dataset compared with those of Models 2 and 3, respectively. The accuracy of the proposed model was 0.19 % lower than that of Model 1 on the B3 dataset. AdaptMLP can improve the feature extraction details of the model. However, in a specific B3 dataset, focusing too much on feature extraction may lead to feature distortion, which affects the final classification feature mapping and results in lower model accuracy. In conclusion, each module in the AViTK model is positively correlated with performance.

5. Conclusions and future work

This study developed an innovative ViT network architecture for tool wear state recognition on the basis of a 2D MTF image encoding technique. In the framework of the ViT, a 2D image classification model based on MTF-AViTK was proposed. The conclusions of this study are as follows.

(1) An intelligent tool wear state recognition model based on MTF-AViTK was designed. An MTF, a 2D time series encoding technique, was utilized for the 2D mapping of tool wear and force signals. Token position encoding and self-attention feature extraction were performed for the tool wear feature information contained in MTF images by Adapt-ViT/L 32. Afterward, a KAN was employed to extract class tokens, which corresponded to each class of features for classification purposes. The performance of MTF-AViTK was better than that of most of the existing study. Meanwhile, both the single-image conversion efficiency of the MTF image encoding technique and the single classification time of the model can reach millisecond level in this device. The metric of inference time of the proposed model is guaranteed.

(2) The MTF-AViTK network was validated to perform outstandingly in terms of tool wear classification, with high robustness on the PHM2010 dataset. Regarding the cross-validation datasets, the average classification accuracies were 94.63 % on the B datasets and 92.48 % on the A dataset. The accuracy attained on the A3 dataset was 4.68 % higher than the best performance among the existing models. The accuracy achieved on the B1 dataset was 1.03 % higher than the best performance among the existing models. It was verified that MTF-AViTK model exhibited better generalizability and robustness.

(3) Ablation experiments showed that both AdaptMLP and the KAN contributed to improving the accuracy of the model classification results. Moreover, in terms of tool wear classification, the AViTK model could learn the key feature information contained in the MTF images. Compared with the baseline network, a significant advantage is provided by adopting KAN for feature classification and mapping, which contributes to the recognition accuracy improvement.

The proposed tool wear state recognition framework could accurately identify tool wear states. However, restricted by time and cost issues, the original input signal was a single force signal in this study. Multidimensional signals can more accurately reflect tool wear states. In future work, multisource data fusion experiments for tool wear state recognition tasks with complex conditions could be investigated to exploit the MTF-AViTK model. Moreover, the interpretability and structure of AViTK will be further optimized so that the model can be applied in practical machining scenarios. The potential of the model will be further exploited in practical machining.

CRediT authorship contribution statement

Shengming Dong: Writing – original draft. **Yue Meng:** Methodology. **Shubin Yin:** Formal analysis. **Xianli Liu:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ymssp.2025.112473>.

Data availability

Data will be made available on request.

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